Acoustic systems（1st class）

Yoshikazu Koike
Department of Electronic Engineering

- Overview of "Acoustic Systems" class
- History of Electro-acoustics

Self-introduction (1)

Yoshikazu Koike

Professor, Department of Electronic Engineering

Research field: sensor and actuators

- Piezoelectric sensor and actuators
- Ultrasonic applications
- Underwater engineering
- GNSS applications …

Our developed underwater observation system



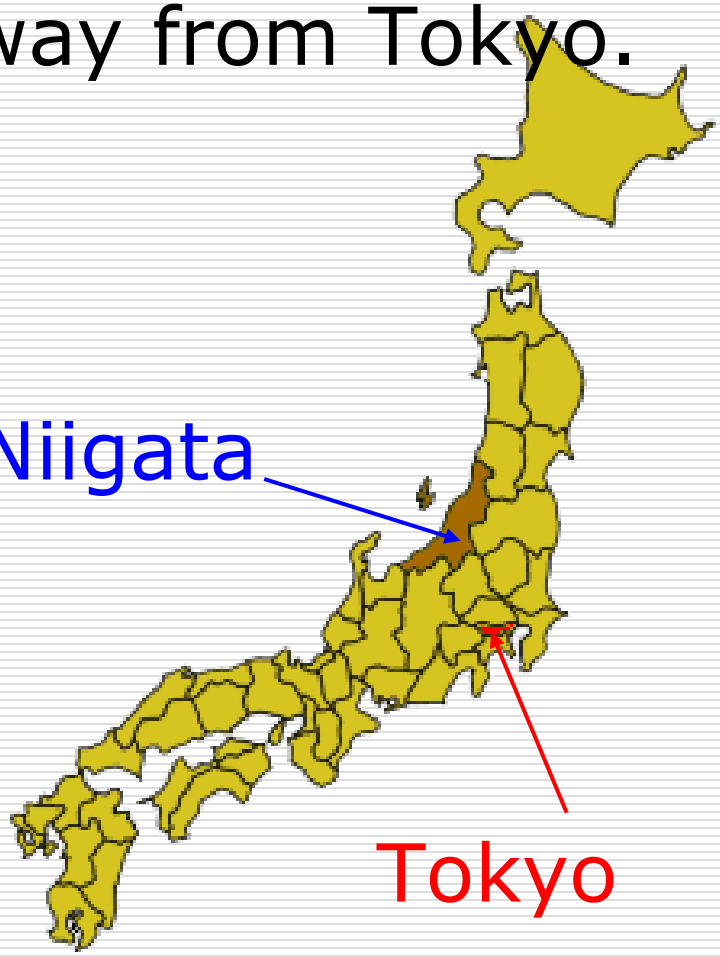
Self-introduction (2)

I am from Nagaoka in Niigata Prefecture.
Nagaoka is about 300km away from Tokyo.

Fireworks festival in Nagaoka
is most famous in Japan

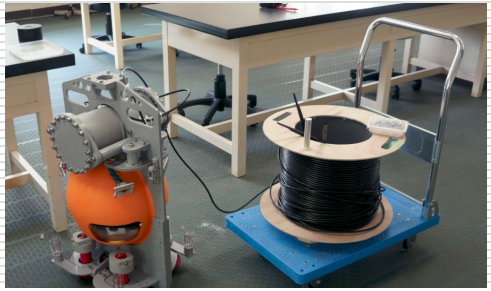


http://nagaokamatsuri.com/zaidans.html



Our laboratory objectives and research field

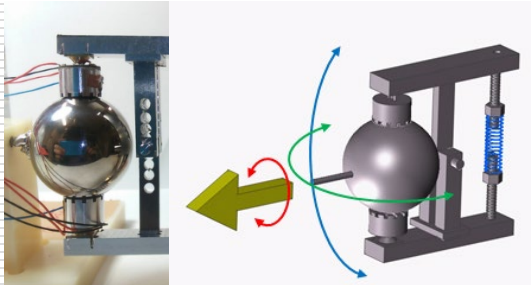
- Contribution to human life by wave information and energy.
- Most of topics are related on the application of sounds and vibration. However, the research field extends beyond sound applications.



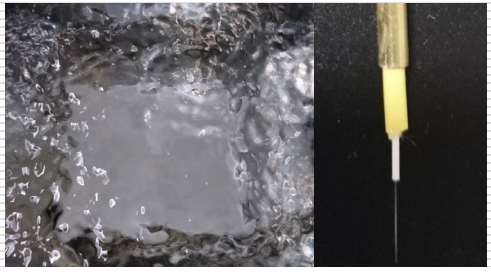
Underwater observation



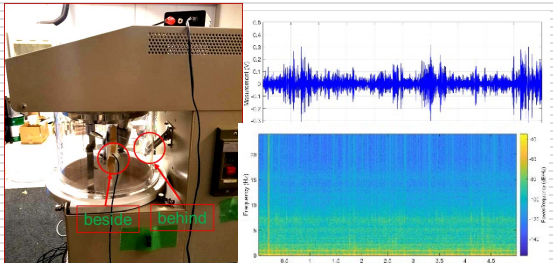
Sound source localization



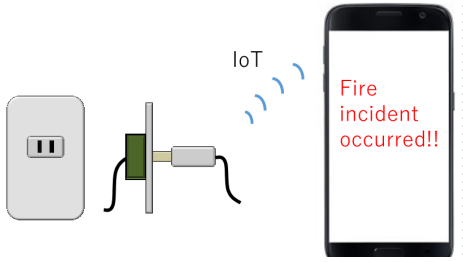
Piezoelectric actuator



Sound measurement by optical fiber



Estimation of food processing progress by sound



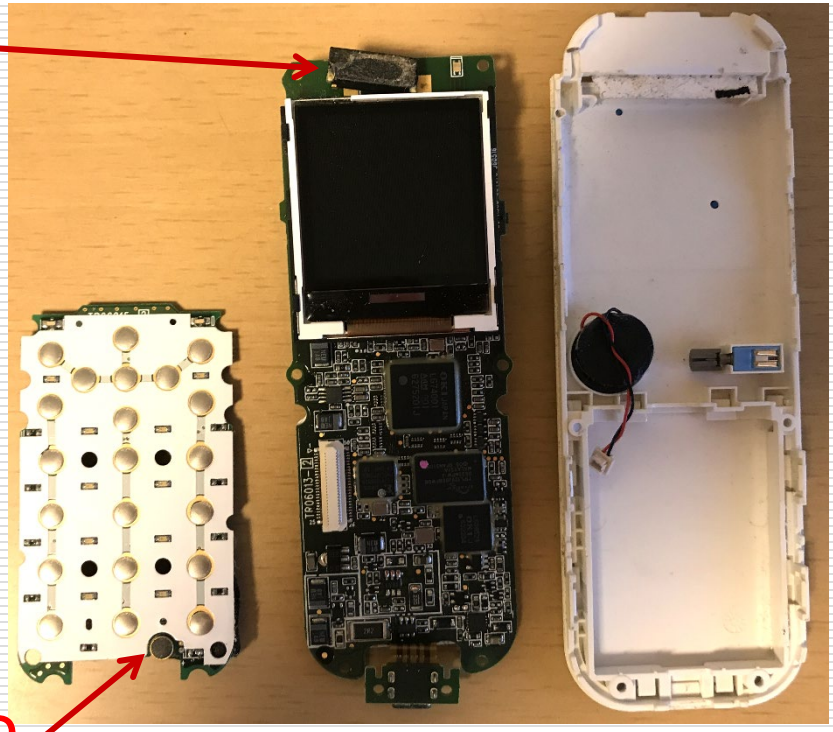
Detection of fire incident inside a plug

Overview of "Acoustic systems" class

An acoustic apparatus is very familiar.

- Everyone has some audio apparatuses.





Design of music hall

- In order to design a music hall, acoustic technique is very important.

Yokohama Minato Mirai Hall



For the acoustic design of the two halls, computer simulation based on geometrical acoustics and a 1/10 scale model experiment were performed for checking room shape and auralization studies.

https://mmh.yafjp.org/mmh/about/main.php

Fields of Acoustics (1)

Research Field	Topics
Speech	Speech production, Speech perception Speech analysis, Speech synthesis Speaker recognition •••
Psychological and Physiological Acoustics	Physiological and neurological mechanism of hearing, Psychology of hearing Auditory model •••
Noise and Vibration	Generation mechanism of noise and vibration Theoretical analysis of transmission of noise and vibration Prediction model of noise and vibration •••
Architectural Acoustics	Analysis for mechanism of vibration and noise Theoretical analysis of vibration and noise Sound insulation measures for buildings •••
Electro－acoustics	Transducers, Measurement and evaluation methods, Sound field recording, analysis •••

Fields of Acoustics (2)

Research Field	Topics
Musical Acoustics	Sound generation mechanism of instruments Psychological property of musical sounds Structure of music and performance •••
Ultrasonics	Ultrasonic properties of materials, phonon physics, acousto-optics, Measurement techniques, imaging, nondestructive evaluation Piezoelectric devices •••
Acoustic Imaging	Imaging by sound wave, Visualization and sonification of sound and vibration Image processing •••

From Acoustical Society of Japan
(https://acoustics.jp/en/overview/research_field_e/)

Objects of the class

- Be able to understand propagation sound and to calculate sound field.
- Be able to understand operation of electro-acoustic systems and to design the systems.
- Be able to understand sound unit associated with the sense of hearing.
- Be able to design sound field using finite element analysis.

Outline of the class

- The class is planned to be 14 times in the semester.
- The class is lectured by slides.
- In the class, I'll submit several questions.
- After the class, upload the image or the PDF file of the answered sheet to ScombZ. I employ the uploaded answer sheet to check the attendance.

Evaluation

4 times exercise or report (22% X 4 =88%)

Final report of acoustic element by computer simulation (12%)

Pass the class when total score is beyond 60%

Bonus subjects

One or two questions are prepared in a class.

If you submit your answer sheet to ScombZ, 2% is added to your score in each time.

Summarize your answer in A4 paper. You may use application like MS- Word. You have to convert your file to PDF.

Schedule of the class (1)

Apr. 15, 1. Introductions of the course, history, the relation of the electronic on the acoustics

Apr. 22, 2. Propagation of the sound in one dimensional system

Apr. 29, 3. Plane wave propagation and wave equation of the sound in three dimensional system (1st exercise)

May 13, 4. Acoustic impedance and acoustic absorption

Schedule of the class (2)

May 20, 5. Basic principle of the vibration

May 27, 6. Analogies of acoustics to machineries and electronics (2nd exercise)

June 3, 7. Kinds of an electro-acoustic transducer and an operation equation of the transducer

June 10, 8. Design of principle of an electro-acoustic transducer (1)

Schedule of the class (3)

June 17, 9. Design of principle of an electro-acoustic transducer (2)

June 24, 10. Design of principle of an electro-acoustic transducer (3) (3rd exercise)

July 1, 11. Hearing of sense

July 8, 12. Digital signal processing for acoustics (4th excercise)

Schedule of the class (4)

July 15, 13. Practice of sound filed calculation using FEM (1)

(Practice using FEM application at room 604)

July 22, 14. Practice of sound filed calculation using FEM (2)

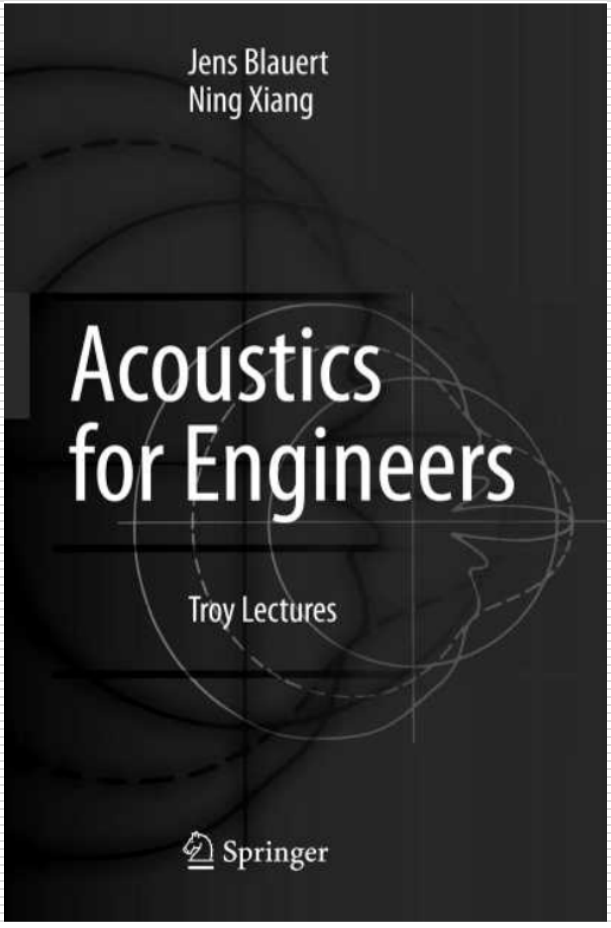
(Practice using FEM application at room 604)
(FEM report assignment)

Reference material

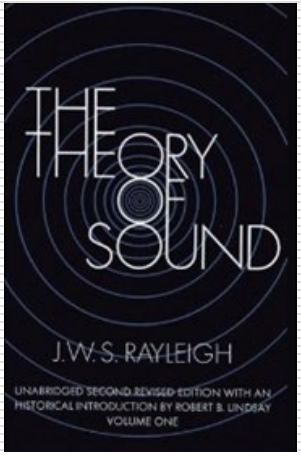
Jens Blauert, Ning Xiang:

Acoustics for Engineers,

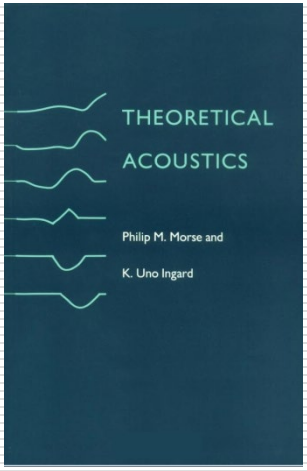
Springer Publishing.



Another famous Reference material



Lord Rayleigh:
Theory of sound
(This book is available on Web)



P.M.Morse and K.U.Ingard:
Theoretical Acoustics

Q1:Please write your motivation to take "Acoustic systems" class.

History of Electroacoustics

Acoustics and Electronics:

Classic acoustics

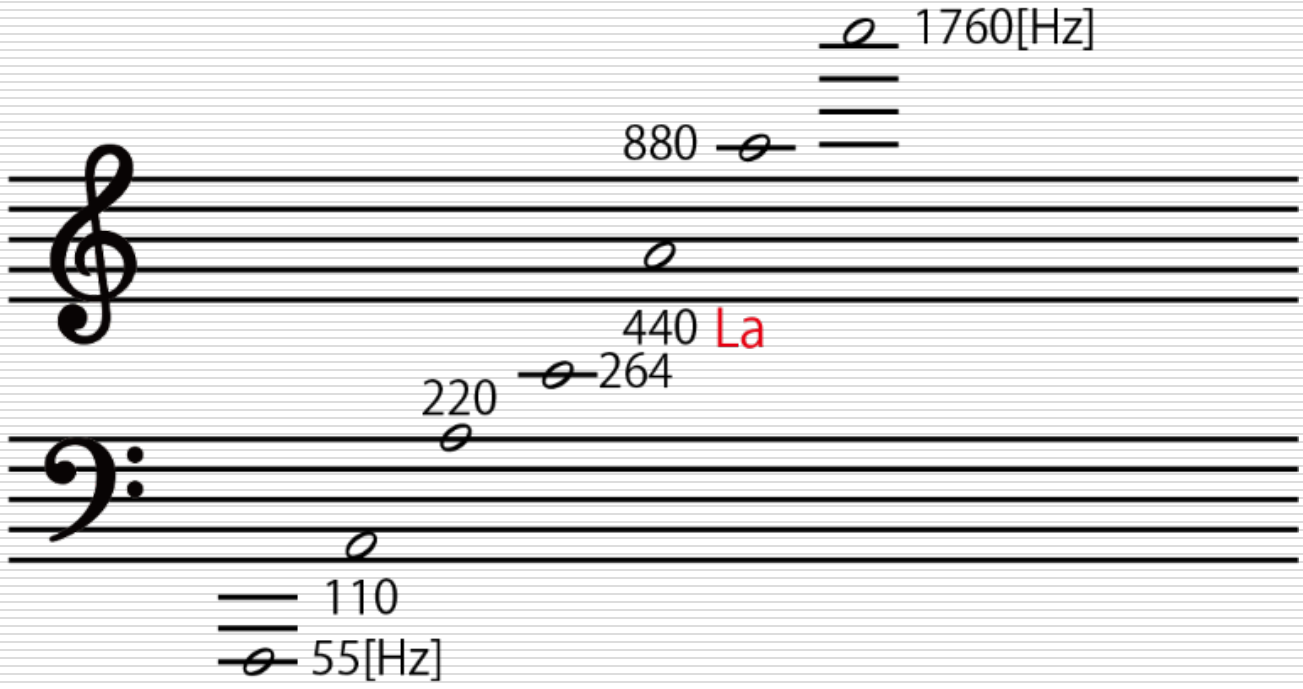
Pythagoras (500BC)
Regularity of sound



Evangelista Torricelli (1643):
No sound is in vacuum
Robert Boyle (1660):
Propagation of sound requires air.
Hermann Helmhortz (1824-1891)
J.W. Baron Rayliegh (1842-1919):
Systemization of acoustics

Concepts of sound

Pythagorean temperament was born in around B.C.400.



Acoustics and Electronics:

Alexander Graham Bell (1847-1922):
Telephone
Thomas Alva Edison (1847-1931):
Carbon microphone, gramophone

Electrical acoustics



Lee de Forest (1873-1961):
Triode vacuum tube (1906)

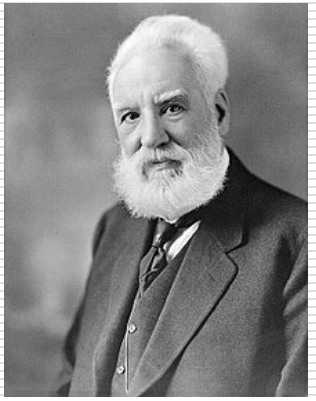
Electronic acoustics

The modern acoustics began.

Bell invented telephone

Alexander Graham Bell:

Bell considered that voice can be sent at light speed if variation of sound can be converted to variation of current.



https://en.wikipedia.org/wiki/Alexander_Graham_Bell

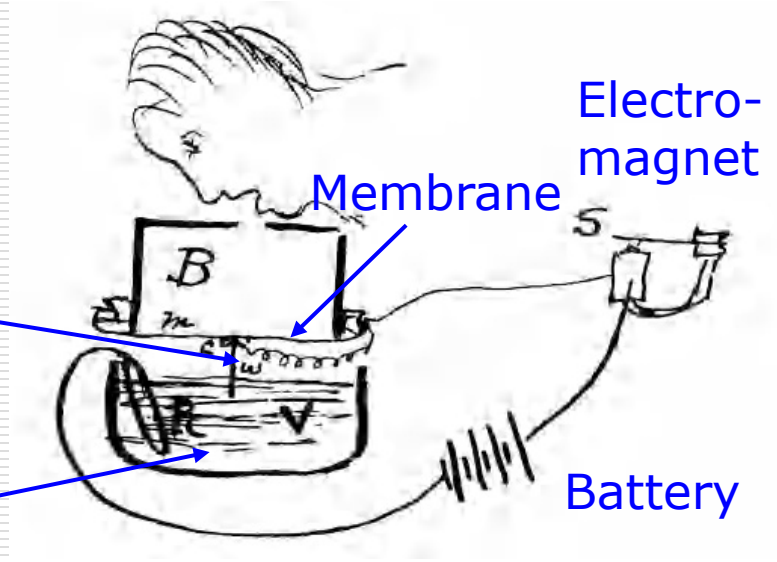
Bell obtained the patent of the telephone at March 7, 1876.

After 10 days from the innovation of the telephone, he succeeded the conversion of voice to current signal.



From NTT Facebook page

Sketch of Bell innovation



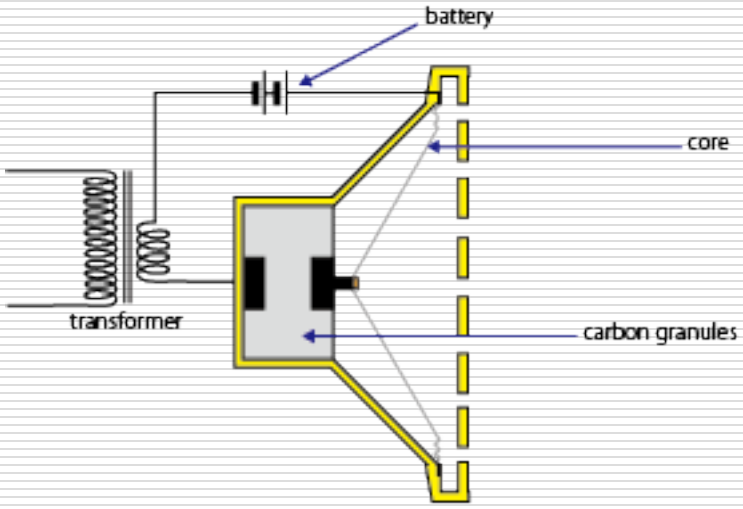
http://telephonecollecting.org/articles/bellinvent2.html

Q2:Explain the sketch.

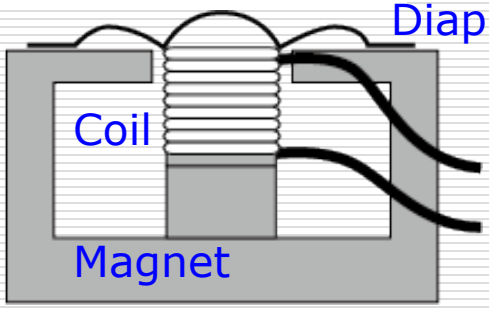
Carbon microphone: One of the most significant devices Edison innovated.

Typical microphone employed currently

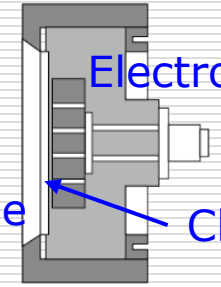
Carbon microphone



Dynamic microphone



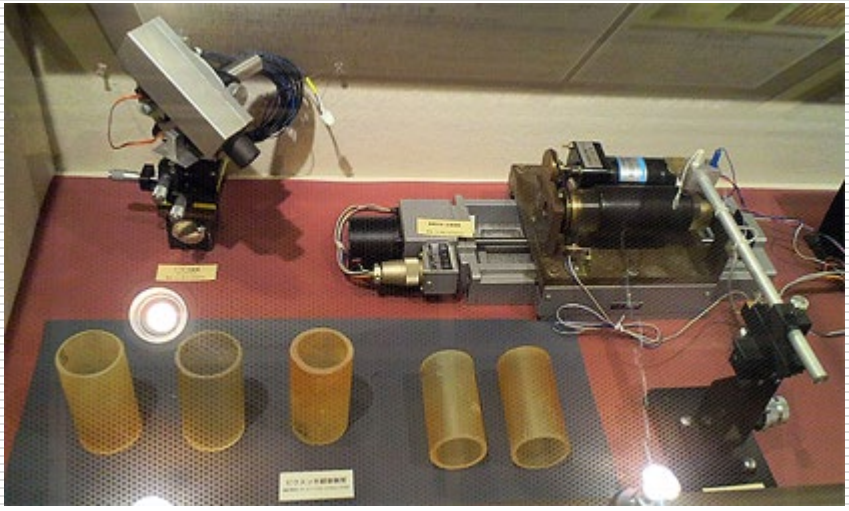
Condenser microphone



Carbon microphone improved greatly the ability of the telephone bell innovated.

Gramophone (or Phonograph)

Wax tube gramophone recorder



http://www.consadole.net/kariki/article/1268

Wax tube gramophone player

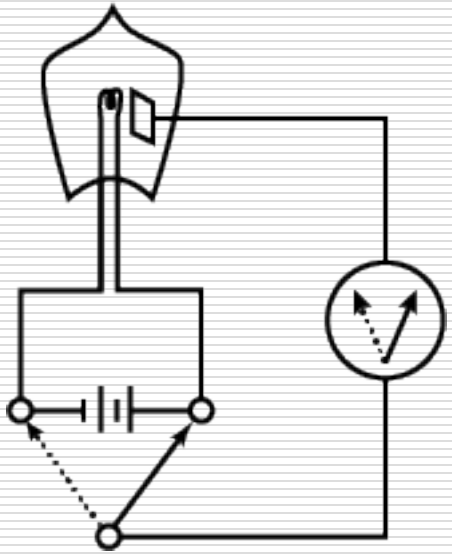


Edison effect and Innovation of Vacuum tube

Edison discovered at 1884.

Current was detected by galvanometer when the filament connected to anode.

This phenomena is called as Edison effect.

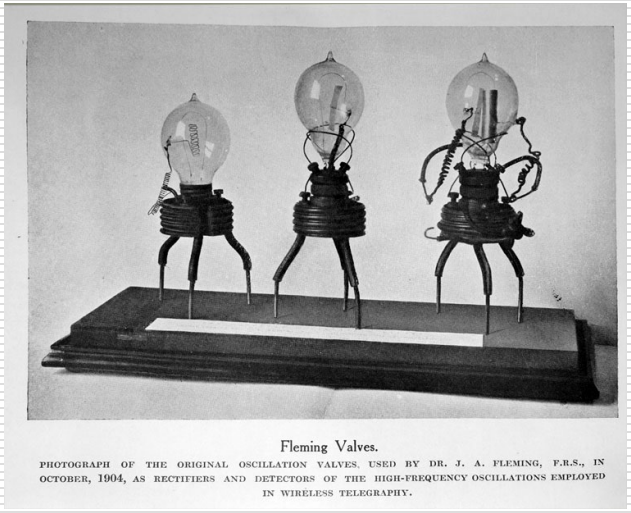


Edison effect is regarded as a trigger for the innovation of diode tube.

Innovation of vacuum tube

A Bipolar vacuum tube was innovated by Fleming at 1904. He applied the vacuum tube to a detector.

A bipolar vacuum tube plays the same role as a diode.



Fleming Valves.

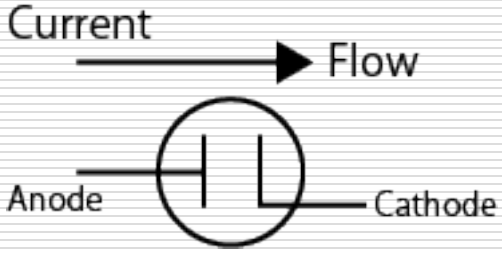
PHOTOGRAPH OF THE ORIGINAL OSCILLATION VALVES, USED BY DR. J. A. FLEMING, F.R.S., IN OCTOBER, 1904, AS RECTIFIERS AND DETECTORS OF THE HIGH-FREQUENCY OSCILLATIONS EMPLOYED IN WIRELESS TELEGRAPHY.

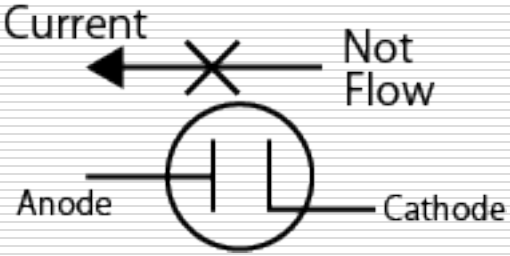
https://www.gracesguide.co.uk/File:Im1917YBWT-Fleming2.jpg

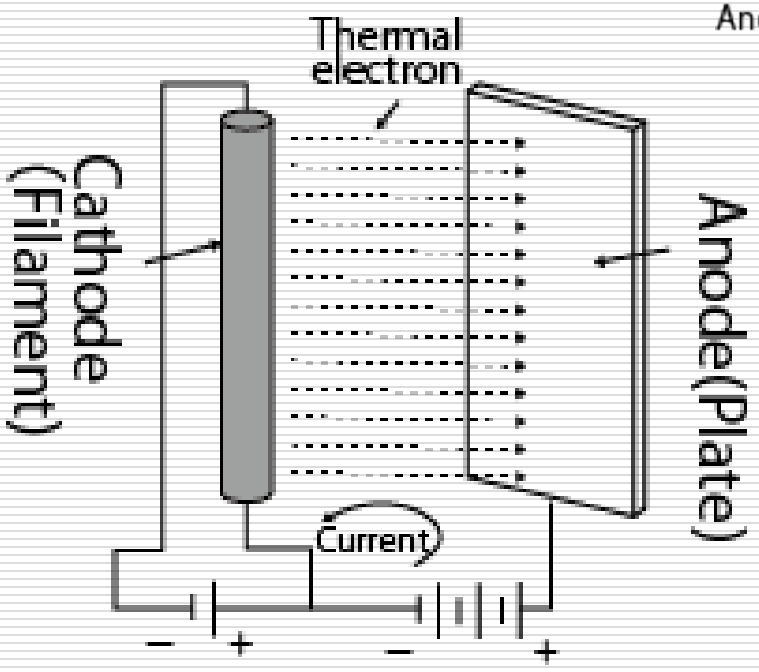
Structure and behavior of a bipolar tube

Bipolar tube was employed as a detector.

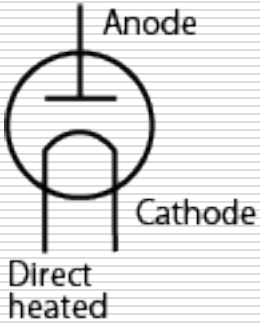
Thermionic emission

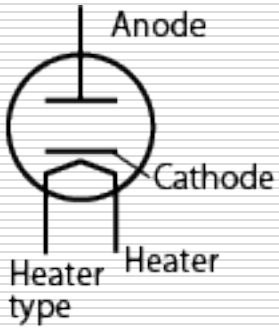






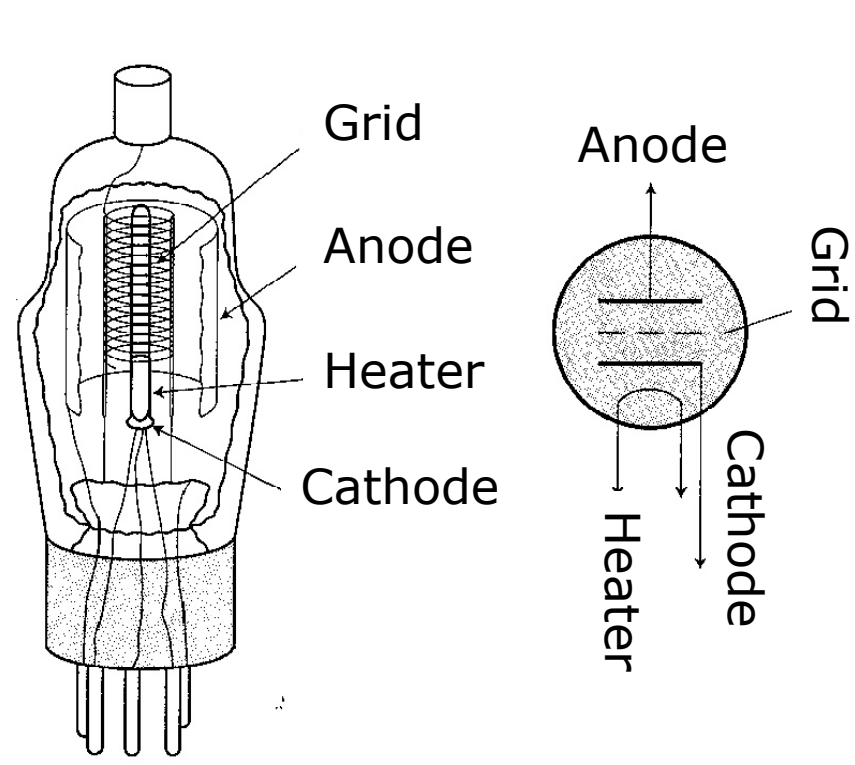
Structure for Thermionic emission



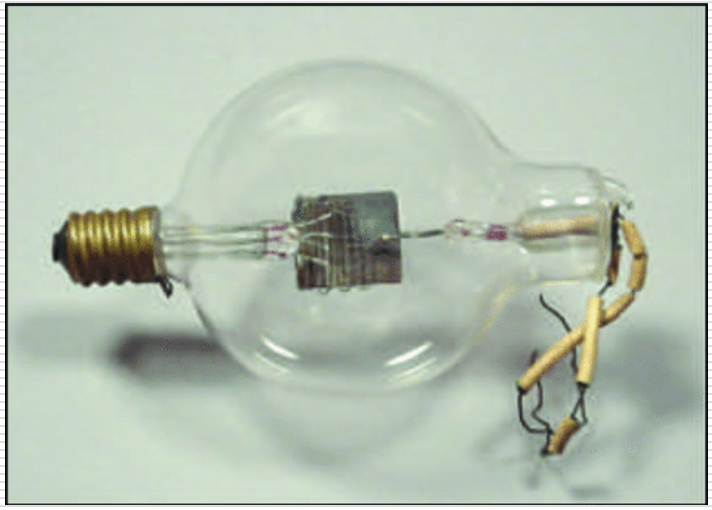


Innovation of a triode vacuum tube

De Forest innovated triode vacuum tube.



Early era triode



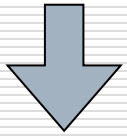
https://www.researchgate.net/figure/An-early-model-of-De-Forests-Audion-Image-courtesy-of-Reverse-Time-Page-at_fig2_254057589

From vacuum tube to transistor, IC

Life time of a vacuum tube is too short. And, also volume is very large.



Main electronic parts for amplification became transistor.



Now a days, integrated circuit (IC) is employed ordinary owing to digital signal process, information theory.